

Milestone 2

Geospatial & Transit Analytics

WEEKS 3 – 4

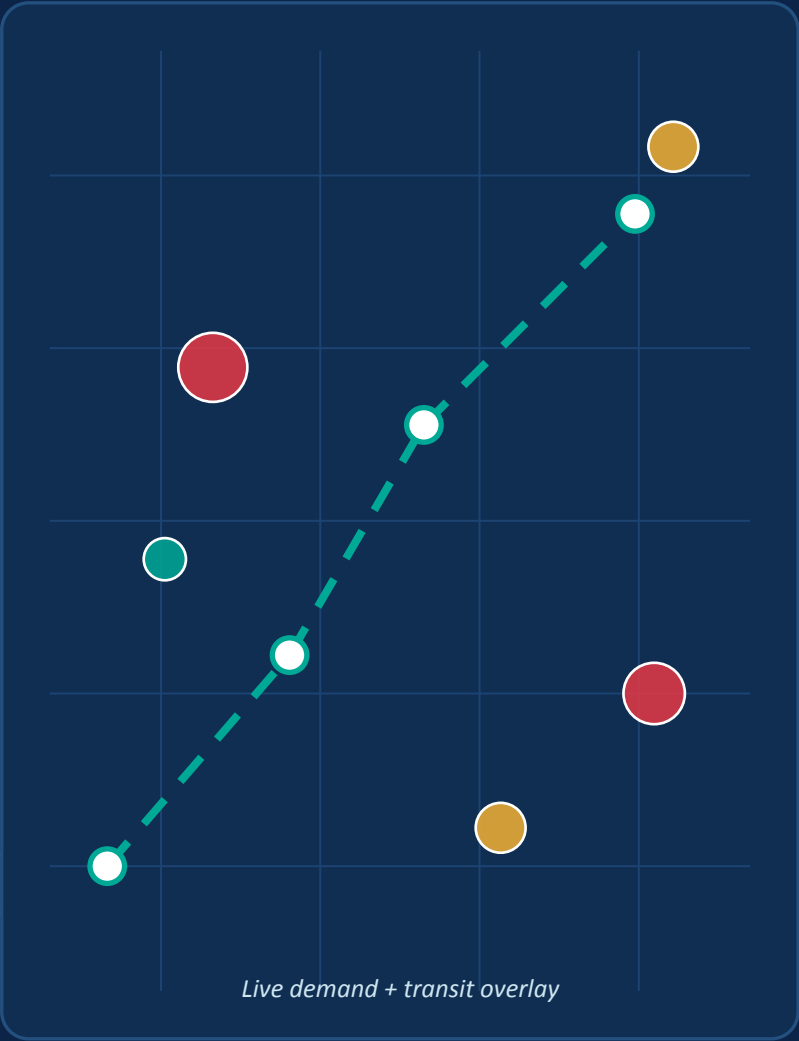
OBJECTIVE

Understand where mobility demand occurs across the city and how transit services perform against it — connecting geographic demand patterns with real-world service reliability.

Demand Heatmaps

Origin–Destination Flows

Transit Performance KPIs



Mobility Demand Heatmap

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Zone-level demand intensity mapped across Chennai using latitude / longitude trip data

HOW WE MEASURE DEMAND

- **Demand by zone** Trip requests aggregated into geographic zones using rider pickup latitude / longitude.
- **Intensity tiers** Zones classified as High, Medium, or Low demand based on trip-volume density.
- **Geographic drill-down** Analysts can zoom from city-wide view down to a single zone or street segment.
- **Chennai hotspots** T. Nagar, Anna Nagar and OMR IT corridor emerge as the highest-demand zones.

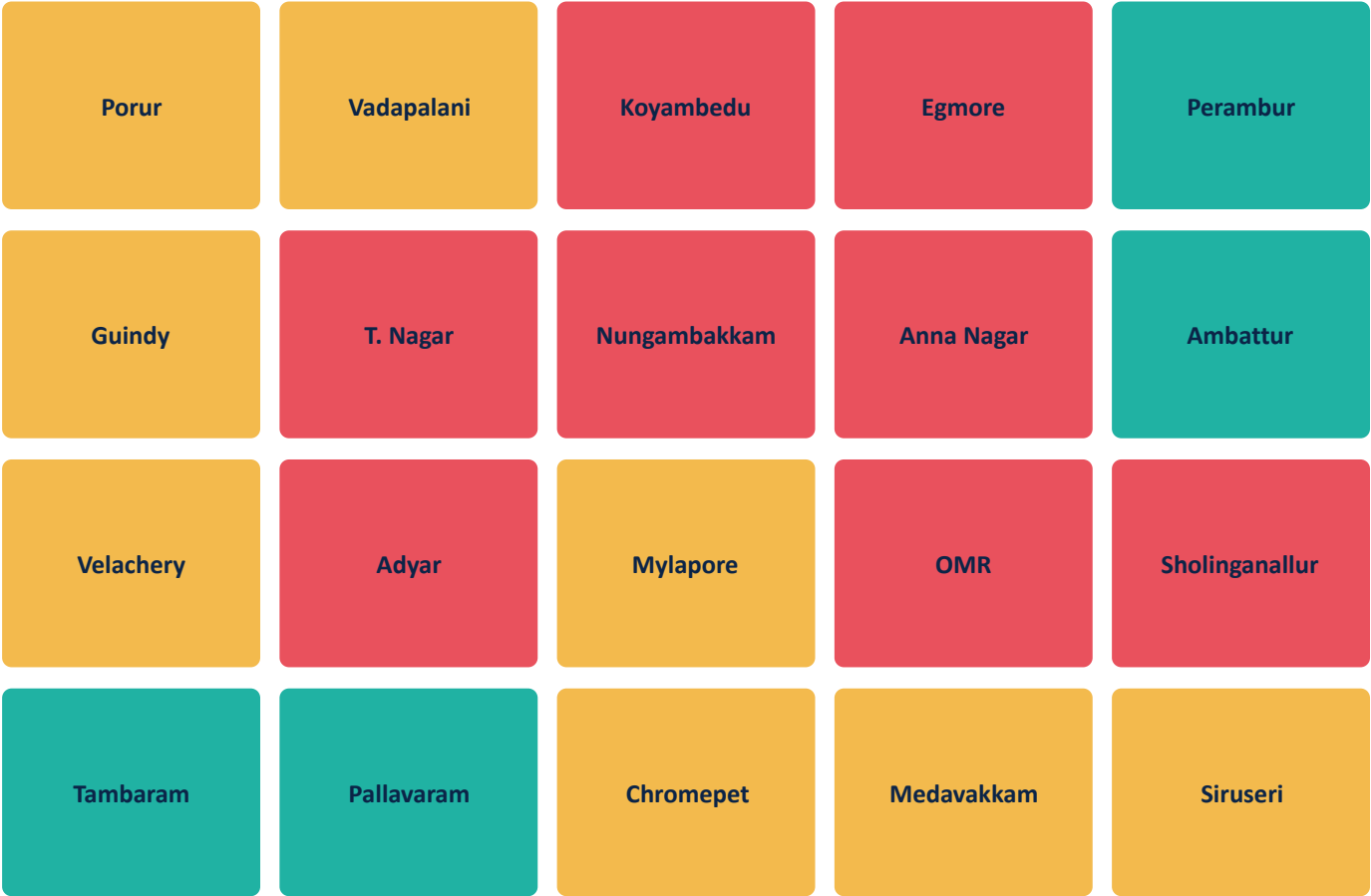
DEMAND LEGEND

High demand

Medium demand

Low demand

Chennai Demand Heatmap — Sample Zones



Grid cells approximate lat/long-binned zones; color = relative trip-demand density.

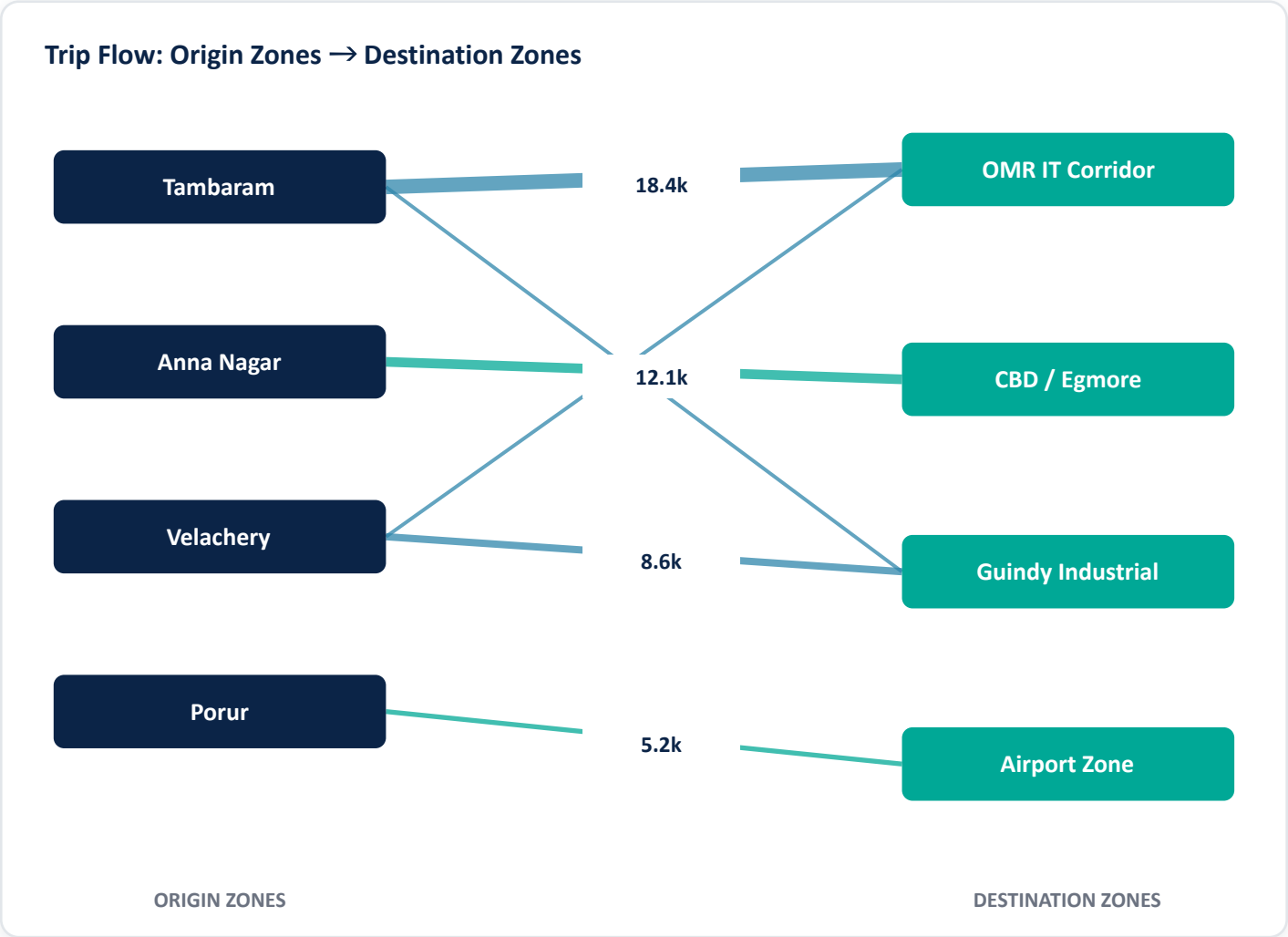
Origin–Destination Flow Analysis

Mapping major mobility corridors between origin and destination zones

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- **Origin zones** Residential-heavy areas where trips typically begin (e.g. Tambaram, Anna Nagar).
- **Destination zones** Employment / commercial hubs where trips end (e.g. OMR IT Corridor, CBD).
- **Major corridors** High-volume paths repeatedly traveled between origin and destination pairs.
- **Trip flow volume** Line thickness reflects the number of daily trips moving along each corridor.

Top corridor: Tambaram → OMR IT Corridor, carrying ~18,400 trips/day — the single busiest origin–destination pair in the network.



Transit Performance

Power BI-style KPIs for schedule reliability, headway consistency and passenger load

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87.4%

On-Time Performance

91.2%

Schedule Adherence

8.4 min

Average Headway

±2.1 min

Headway Variance

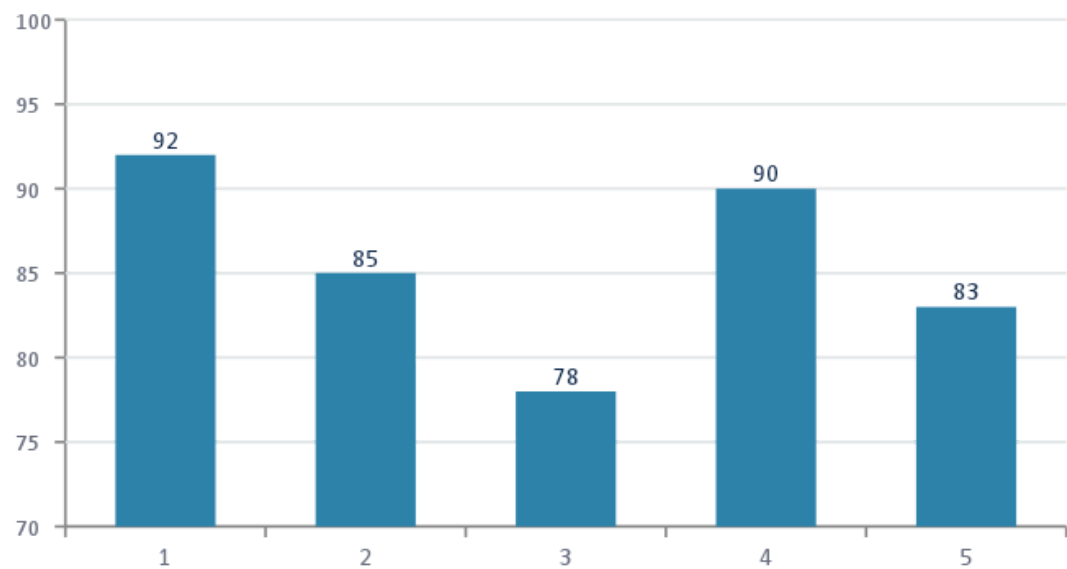
42,500

Daily Passenger
Throughput

78%

Peak Crowding Index

On-Time Performance by Route



Route Performance Summary

Route	On-Time %	Avg Headway	Throughput/Day	Crowding
Route 1 — OMR Express	92%	6.5 min	9,800	High
Route 2 — Anna Nagar Loop	85%	8.2 min	7,300	Medium
Route 3 — Tambaram Link	78%	10.1 min	11,200	High
Route 4 — Adyar Connector	90%	7.4 min	6,900	Medium
Route 5 — Porur Feeder	83%	9.0 min	7,300	Low

Key Insights & Next Steps

1

Demand is concentrated

T. Nagar, Anna Nagar and the OMR IT Corridor account for the largest share of city-wide mobility demand.

2

One corridor dominates

Tambaram → OMR IT Corridor is the busiest origin–destination pair, moving ~18,400 trips/day.

3

Reliability varies by route

On-time performance ranges from 78% to 92% across routes, with Route 3 needing attention.

NEXT: MILESTONE 3

Predictive Demand Forecasting & Route Optimization

- Forecast demand spikes by zone
- Simulate route/schedule adjustments
- Prioritize high-crowding routes for optimization